NPTEL MCQ Revision Notes — Lectures 11–20

Course: Sustainable Transportation Systems | **Topic block:** Environmental Impact Assessment (EIA) | **Instructor shown on slides:** Prof. Bhola Ram Gurjar

Source: only the 10 uploaded PDFs (lec 11.pdf … lec 20.pdf). No outside knowledge added. Tags like L12-s8 = Lecture 12, slide 8. Text in *[brackets]* is a note about the source. Yellow boxes = extraction issues. Slides that were pictures/diagrams/tables were rendered and read; their content is merged into the notes.

Uploaded PPTs

File	Lecture title	Slides
lec 11.pdf	L11: Introduction to Environment Impact Assessment (EIA)	47
lec 12.pdf	L12: EIA Processes – I (project proposal, screening, scoping, baseline, impact assessment)	51
lec 13.pdf	L13: EIA Processes – II (mitigation, impact management, EIS, review, public participation, follow-up)	43
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lec 20.pdf	L20: EIA Case study-IV (Guwahati airport terminal)	36

Page guide: Part 2 = lecture-wise notes · Part 3 = rapid-revision sheet · Part 4 = 30 practice MCQs + answer key · Part 5 = coverage audit.

> **Note:** A few slides in the PDFs are blank in the PDF itself (their text boxes were not saved), e.g. L14 pros/cons slides and L13 s30. These cannot be recovered and are listed in Part 5. Ink marks (red pen) on slides are ignored.

Lecture 11 — Introduction to Environment Impact Assessment (EIA) (47 slides)

Development vs environment; EIA as middle path L11·s3–8

- Choice posed (s3): **Development** (poverty reduction, employment, ease of living) vs **Environment** (environment protection, biodiversity, human health). "Unplanned developments negatively affect the environment" (s4).
- **Aarey Forest (s5):** approx. **800-acre** land in Mumbai; **2600 trees** to be cut for metro shed; protest erupted after **1500** trees were cut; government later changed decision and declared it **reserved forest land**.
- **Kudankulam Nuclear Power Plant (s6):** project to reduce dependence on coal-based thermal plants; villagers protested citing negative environmental impact; example of **regional/national interest vs local interests**.
- **Madhyam-Pratipada (s7):** Buddha — "enlightenment by avoiding the extremes of self-gratification on one hand and self-mortification on the other" = **middle path**; answer to development-vs-environment = **balancing the two sides**. **EIA = balancing process between developmental and environmental goals** (s8).

Definition, need, purpose L11·s9–14

- **EIA:** a tool for decision making to find the best possible alternative in terms of environmental as well as economic costs and benefits.
- **Definition (Barry Sadler 1996):** "a process of **identifying, predicting, evaluating, and mitigating** the **biophysical, social, and other relevant effects** of proposed projects and physical activities **prior to major decisions and commitments** being made."
- **Need (s10):** EIA aims to minimize environmental and social impacts and can enhance sustainability development where large-scale projects are implemented (IISD 2016). s11: impacts of transportation projects — **Economic = generally positive**; **Social = positive/negative**; **Environmental = generally negative**.
- **Purpose (s12):** (1) **facilitate decision-making** – systematic examination of environmental implications of a proposed action and alternatives before a decision; (2) **aid in formation of sustainable development** – framework for considering location, design and environmental issues in parallel; aid to formulation of developmental actions; environmentally sensitive developments; (3) **instrument for sustainable development** – overall quality of life; continuing access to natural resources; avoiding lasting environmental damage.
- **Core values (s13): Integrity** (fair, unbiased, balanced); **Utility** (balanced, credible information for decision making); **Sustainability** (should result in protecting the environment).
- **Components (s14):** 1 Resource capability and ecosystem mapping; 2 Natural hazard and landscape sensitivity evaluation; 3 EIA and other forms of impact assessment; 4 Ecosystem approaches (e.g. critical zone, habitat supply).

Evolution of EIA L11·s15–21

Year	Event (slide 15 timeline)
1960s	EIA as part of a rational decision-making process; a technical evaluation that would lead to decision making
1969	EIA made legislation in the US in the **National Environmental Policy Act (NEPA)**
1980s	**World Bank adopted EIA** for major development projects; **UNEP adopts EIA** studies
1991	**Espoo Convention** – to consider trans-boundary effects in EIA studies
1992	**Rio Declaration** – UN declares EIA as a national decision-making instrument
1994	Union Ministry of Environment & Forests, GOI, under **Environment (Protection) Act 1986**, promulgated EIA notification
1997	**Public hearing was made compulsory** before impact assessment was finalized
2006	MoEFCC notified new EIA legislation
2020	MoEFCC notified new EIA draft

Period	Development of EIA worldwide (IAIA 1996, s16–18)
Pre-1970	Project review based on technical/engineering and economic analysis; limited consideration to environmental consequences

Early/mid-1970s	EIA introduced by NEPA 1969-70 (US). Standard methodologies developed (**matrix, checklist, network**). **Canada, Australia, New Zealand and Columbia** first to follow NEPA in 1973-74; **Australia legislated EIA**, Canada and New Zealand established administrative procedures
Late 1970s–early 1980s	More formalized guidance. Formal EIA requirements: **France 1976, Philippines 1977**; informal/experimental use: **Netherlands 1978**; elements adopted in planning-permission applications: **UK 1988**. Use by developing countries (Brazil, Philippines, China, Indonesia)
Mid-1980s–1990	EC Directive on EIA establishes basic principle and procedural requirements for all member states; efforts to address **cumulative effects**; World Bank and other aid agencies establish EIA requirements; spread in Asia. Also: Strategic Environment Assessment (SEA) and risk analysis included; provision for public involvement; coordination with land use planning
1990s	Requirement to consider **trans-boundary effects** (Espoo); increased use of **GIS and IT**; sustainability principle and global issues; **India adopted EIA formally**

- **Espoo Convention (1991, s19): first multilateral EIA treaty**; treats EIA in a trans-boundary context; **entered into force 1997**; obligations to assess impact of certain activities at an early planning stage; general obligation of states to **notify and consult** each other on major projects likely to have significant adverse environmental impact across borders.
- **Rio Declaration (1992, s20): Principle 17** calls for EIA as a national decision-making instrument for activities likely to have significant adverse impact; emphasized role of **competent national authority**.
- **EIA timeline in India (s21): 1976-77** – Planning Commission asked Dept. of Science and Technology (DST) to examine **river-valley projects** from an environmental angle. **Until 1994** – EC from Central Govt was an **administrative decision and lacked legislative support**. **1994** – EIA notification under EP Act 1986 making EC mandatory. **Sept 2006** – MoEF notified EIA legislation making EC mandatory for various projects; **onus of clearing projects on state government depending on size/capacity** of project. **2020** – new EIA draft notified for public comments.

Types of EIA L11·s22–33

Type	Facts
Rapid EIA (REIA) (s22)	Quick assessment of likely adverse impacts; mostly at initiation of new activity; covers **one season** baseline data for air, noise, water, land, biological and socio-economic components
Comprehensive EIA (s22)	Data for **four seasons** (monsoon, post-monsoon, winter, summer); time taking; conducted only if required and asked by the assessment body
Objective-based classification (s23)	Types based on objectives of project or its theme: **Regional, Sectoral, Strategic, Project level, Life Cycle Assessment**
Life Cycle Assessment (LCA) (s24)	Environmental concerns enter every step; examines impacts of a product at **all stages of product life cycle**; especially manufacturing; considers total picture. Example cycle: Design phase → Raw material acquisition → Materials processing → Construction → Using stage → Maintenance → Recycling
Regional EIA (s25–27)	Environmental concerns integrated with regional planning; integration of economic growth with natural-resource management; critical for SDGs. **Example Doon Valley (Dehradun, Uttarakhand):** industries located per **Water (Prevention and Control of Pollution) Act 1974** and **Air Act 1981**; land use, tourism, grazing per plan approved by MoEF; mining needs prior MoEF clearance; **Red category (most polluting) industries not allowed**
Sectoral EIA (s28–29)	EIA for specific sectors (transportation, energy, mining). **Transportation key features:** wider (national/global) impacts; promotes integration of modes; overall assessment of energy and fuel consumption; encourages sustainable use of resources
Strategic EIA (SEA) (s30)	Systematic analysis of environmental effects of development **policies, plans, programmes (PPP)**; proactive; **"SEA is the EIA of policies, plans and programs"**
Project level EIA (s33)	Activities considered in isolation; specific to proposed project; **most common**; e.g. EIA for a highway section or small project

Difference between EIA and SEA (s31)

Aspect	EIA	SEA
Timing	End of decision-making cycle	Earlier stages of decision-making cycle
Nature	**Reactive** to a development proposal	**Pro-active** to development proposals

Scope	Specific project; specific impacts	Areas, regions or sectors; effects of policy/plan/programme on environment
Alternatives	Limited number of feasible alternatives	Broad range of potential alternatives
Emphasis	Mitigating and minimizing impacts	Meeting environmental objectives, maintaining environmental quality
Perspective	Narrow, high level of detail	Broad, lower level of detail (vision and overall framework)
Process	Well-defined, clear beginning and end	Multi-stage, overlapping components; policy level continuing and iterative
Framework	Project-specific impacts	Creates framework against which impacts and benefits can be measured

- **SEIA example – Helsinki Metropolitan Transport System (s32): 3 alternatives** compared: existing 1994 system (baseline); policy to develop public transport; policy to develop private vehicle traffic. Positive/negative points identified on **land use, environment, traffic and mobility**.

Classification of impacts L11·s34–39

- s34 (Bai Yong & Bai Qiang 2014): **Environmental burden** (discharge to sea; emission to air; waste to land) → **Ecological impact** (ecological functions, biodiversity, habitats, endangered species, disease vectors, agriculture); **Social impact** (health, safety, recreation, culture heritage, perceptions); **Economic impact** (revenue/costs; NPV/internal rate of return; employment; community welfare; social equity; national economy). Impacts are **direct, indirect, secondary, and cumulative**; also contribution from other activities in area under assessment.

Classification	Example on slide
Beneficial or detrimental (s35)	Road construction/widening: detrimental = deteriorating air quality; beneficial = improved air quality by using roads for public transport and NMT
Naturally reversible or irreversible (s36)	Water bodies: reversible = maintaining natural flows enhances self-purification (increased dissolved oxygen); irreversible = extensive pollution leads to extinction of species
Repairable or irreparable (s37)	Deforestation for road: repairable = forest cover increased by plantation; irreparable = once sensitive ecological cycle is disturbed
Short-term or long-term (s38)	Metro project: short-term = congestion on roads during construction; long-term = GHG emissions over project life
Local / regional / national (s39)	Local = dirt from unpaved roads and construction sites; Regional socio-economic = benefits of BRTS/MRTS; Regional environmental = acid rain due to SOx/NOx; National = alternate fuels (implementation of **BS VI**) in India

Forms of impact assessment; benefits; flaws L11·s40–44

- **Forms (s40):** Social Impact Assessment; Risk Assessment; Life Cycle Analysis (LCA); Energy Analysis; Health Impact Assessment; Regulatory Impact Assessment; Species Impact Assessment; Technology Assessment; Economic Assessment; Cumulative Impact Assessment; Strategic Environmental Assessment (SEA); Integrated Impact Assessment. An EIA study can include any of these depending on project type and need.
- **Benefits of EIA (s41):** Transparency · Certainty · Participation · Practicability · Flexibility · Cost effectiveness · Credibility · Accountability.

Benefit of EIA (IISD 2016 / Sadler 1996, s42)	Very	Moderately	Marginally	Not
Including a full range of considerations (social, ecological, risk, etc.)	11%	53%	27%	6%
Making precise, verifiable predictions	2%	34%	45%	15%
Identifying appropriate mitigation measures	12%	57%	24%	3%
Indicating confidence levels for data used in predicting impacts	3%	17%	43%	32%
Specifying significance of residual impacts	3%	28%	45%	19%
Providing clear information for decision makers on potential consequences	14%	51%	26%	8%
Providing relevant advice on alternatives to the proposal	8%	34%	37%	16%

Benefits (s43)	Flaws (s43)
Systematic method of impact assessment	Time consuming
Estimates cost/benefit trade-off of alternate actions	Costly
Facilitates public participation	Little public participation in actual implementation

Effective mechanism for coordination, environmental mitigation, negotiations, feedback, top-level decision making	Unavailability of reliable data (mostly in developing countries)
Balance between impact of developments and environmental concerns	Too focused on scientific analysis (sometimes); compliance monitoring after EIA seldom carried out

- **Externalities (s44): Positive** – new jobs and economic growth; growth of local business enterprises; supporting/complementary industries; influx of capital and disposable income. **Negative** – additional pressure on physical infrastructure; escalation of cost; extended time in project implementation.
- **Conclusion (s45):** EIA = effort for harmony with nature without compromising economic benefits; considers impacts by type, level, nature; prevents damage and helps long-term developmental/economic profits → sustainable development.

Important Distinctions to Remember

- **Rapid EIA** (one season) vs **Comprehensive EIA** (four seasons).
- **EIA** (end of cycle, reactive, project) vs **SEA** (early, pro-active, policies/plans/programmes).
- Impact pairs: beneficial/detrimental; reversible/irreversible; repairable/irreparable; short/long term; local/regional/national.
- Espoo 1991 (first multilateral EIA treaty; in force 1997) vs Rio 1992 (Principle 17). NEPA 1969 (US) vs India 1994 / 2006 / 2020 draft; public hearing compulsory 1997.

Image slides read: s10, s11, s15, s21, s24, s34, s39, s42. s4 is a photo (Delhigreens). s26 (map) captions only.

Lecture 12 — EIA Processes-I (51 slides)

Stages L12·s3–6

- **Stages of EIA (s3):** Project Proposal → Screening → Scoping → Baseline Data → Impact Assessment (**this lecture**) → Mitigation → Impact Statement → Review → Follow-up (**next lecture**).
- **Flow chart (s4, Kalita 2016):** Proposal identification → Screening → (EIA required / initial environmental examination / No EIA) → Scoping → Impact analysis → Mitigation and impact management → EIA report → Review → Decision-making → Approved (Implementation and follow up) or Not approved (Redesign → Resubmit). **Public involvement** typically at scoping and review (may occur at any stage). Information from follow-up contributes to effective future EIA.
- **EIA phases (s5):** Screening (does project require EIA?) → Baseline studies (establish environmental baseline) → Alternatives (consider different approaches) → Impact prediction (forecast impacts) → Impact assessment (interpreting impacts) → Mitigation (what can be done to alleviate negative impacts) → EIS preparation/review (document EIA findings) → Public consultation (general public and NGOs) → Monitoring.
- **Project life cycle (s6): Phase A Pre-feasibility** (site selection, environmental screening, initial assessment, scoping of significant issues) → **Phase B Feasibility** (detailed assessment of significant impacts, identification of mitigation needs, input to cost/benefit analysis) → **Phase C Design and engineering** (detailed design of mitigation measures) → **Phase D Implementation** (implementation of mitigation measures and environmental management strategy) → **Phase E Monitoring and evaluation** (monitoring, post-auditing, lessons for future projects) → back to project concept.

Project proposal; clearance L12·s7–10

- Define project and related activities; specify **zone of impact (direct and indirect)**; understand activities closely related to project.
- **Environmental clearance process (s8):** Start → Site selection → Conduct EIA → Apply for **NOC** → **SPCB arranges public hearing** → Project proponent applies for environmental clearance with EIA report, NOC from SPCB, etc. → **Review by Environmental Appraisal Committee** → Accepted / Rejected / Change suggested (loop back to conduct EIA).
- **Zones (s9–10):** Net project area ⊂ Total project area ⊂ Area of direct influence. Highway: **net project area = right of way**; **total project area = area up to buildings**; **area of direct influence = area affected by noise, air pollution or otherwise**.

Screening L12·s11–27

- **Decide whether to conduct the assessment** (likely significant impacts) **or not** (anticipated absence of such impacts); must follow procedures often described in legislation (local laws).
- **Flowchart (s12):** screening identifies environmental issues; determines whether EIA needed; establishes need for project → **Not require EIA / Require EIA / Impacts unclear**. **Threshold criteria:** size, location, output, cost/finances, environmental effects etc. **Impact criteria:** significant but easily identifiable impacts; significant impacts; sensitive area. (Project → environmental, economic, social impact.)
- s13: oil extraction, mining or bullet-train project will have significant impacts; s14: Mussoorie tourist project example (increased vehicular traffic, air traffic, visitors/hospitality, commercial traffic). Screening is done in a predetermined manner to ease the complex process.
- **Parameters (s15):** type of project (mining, river training, transport etc.), size of project (cost, length of road, project area), potential impacts (long/short term, moderate/severe, reversible/irreversible).
- **Standardized approach (s16–17):** 1 activities outlined in legislation and regulations; 2 legal/policy definitions of proposals; 3 **inclusion lists** (with or without thresholds); 4 **exclusion lists** (insignificant or exempt by law e.g. national security/emergency). Examples: all nuclear-substance activities → compulsory EIA (e.g. nuclear power plant); projects needing resettlement (e.g. reservoir); **four-or-more-lane highway > 100 km**; **border infrastructure exempted** in India.
- **Customized approach (s18–21):** case-by-case with indicative guidance; categories (A, B, C or 1, 2, 3). **High impact:** significant adverse impacts (irreversible damage, vulnerable ethnic minorities, involuntary resettlement, cultural heritage sites) e.g. large industrial plants, reclamation/resettlement/new land development, thermal and hydropower, pesticides/hazardous materials. **Medium impact:** limited EIA to identify mitigation – e.g. rural electrification transmission, small-scale aquaculture/irrigation/drainage, renewable energy, tourism, rural water

supply/sanitation. **Low impact:** minimal/no impacts; **no EIA required** – e.g. small buildings, community garden, community wells, outdoor recreation.

Agency	Screening approach (s22–27)
World Bank **A**	Categories depend on **type, location, sensitivity and scale** of project and nature/magnitude of potential environmental impacts
World Bank **B**	Potential adverse impacts on human populations or environmentally important areas
World Bank **C**	Minimal or no adverse impacts; **no further EA action** beyond screening; e.g. small community project or village roads under **PMGSY (Pradhanmantri Gram Sadak Yojana)**
European Union	EIA depending on size, location, activities, impacts: power stations (MW); landfills (volume, tonnes/day, total capacity); roads (km); gas pipelines (km)

- **EU checklist (s26–27), 18 Yes/No questions:** large change in conditions; new features out-of-scale; unusual/complex effect; large area; transboundary impact; many people affected; many receptors of other types; valuable/scarce resources affected; risk of breaching environmental standards; [item 10 not present in extracted text]; protected sites affected; high probability; long time; permanent not temporary; continuous not intermittent; if intermittent, frequent not rare; irreversible; difficult to avoid/reduce/repair/compensate. **"Yes" = EIA required; "No" = no EIA; in case of more "Yes", EIA will be required.** (European Commission 2001)

Scoping L12·s28–32

- **Scoping:** systematic exercise that **establishes the boundaries** of the EIA; sets basis of analyses at each stage; sets baseline and identifies alternatives; reduces risk of including inappropriate or excluding needed components. Important outcomes: area of study; factors for assessment for baseline studies.
- Example: **Char-Dham All Weather Road** (Uttarakhand) – significant regional environmental, economic, social impacts → very large scope boundary.
- **Scoping flowchart (s30) → TOR report:** Project description · Project site (location on map) · Project alternatives (sites, infrastructure, technologies) · Determination of potential impacts · Analysis and evaluation of risks · Formulation of Environmental Management Plan · Mitigation plan · Monitoring plan · Resettlement Action Plan (if relevant) → **TOR (Terms of Reference) report**.
- **TOR (s31)** = roadmap for EIA preparation, covering issues identified in scoping: project description; agencies/ministries overseeing EIA and deciding; geographic area (**impact zone**); EIA requirements in laws; impacts/issues to study; mitigation and/or monitoring systems; provisions for public involvement; key stakeholders; timeframe; expected work product/deliverables; budget.
- **Methods of scoping (s32, Bai & Bai 2014):** making a plan for public involvement → assembling relevant information → distribute information to affected interest → evaluating significance of issues → developing a strategy for addressing issues → establishing priorities for environmental assessment → identifying the major issues.

Baseline; impact prediction/assessment L12·s33–48

- **Baseline study:** collect all relevant information on **current status of the environment**; sound basis for design/evaluation of post-EIA studies (e.g. early-stage monitoring); decide a timeline for predictions (e.g. number of years of project operations).
- **Impact prediction:** forecasting likely changes in the environment; comparing baseline data with alternatives is done under impact prediction; impacts change over time – assessed over lifetime: **construction** → **operation** → **even after closure** (e.g. nuclear power plants).
- **Impact assessment (s35):** based on key environmental, social, cultural characteristics; itemized list of impacts incl. cumulative effects, social impacts, health risks; **direct, indirect and cumulative**; leads to detailed list of mitigation actions. Example: Ahmedabad–Mumbai bullet train.
- **Environmental factors (s37–40): Land use and soil** (soil removal and pollution; land-cover change: wetlands, floodplains, mangroves, subsoil; surface; topography – altitude, gradients, relief, orientation); example Sundarbans mangroves. **Water** (sewage release treated/untreated; groundwater – aquifers, recharge areas, level of use; surface water – areas affected, use, drainage; floods, sedimentation, erosion, eutrophication). **Air** (emissions: noise, odor, dust; climate factors: rainfall, temperature, radiation, fog, wind; sources of pollutants; fragile areas). **Biodiversity** (endangered species; trees and canopy, migratory birds, indicator species, invasive species).

- **Socio-cultural (s41):** e.g. tribes of Andaman & Nicobar preserved by not allowing external interventions. **Socio-economic (s42):** present land use and zoning; land-use plans (master plans); future trends/development pressures; resettlements (Sardar Sarovar dam – mass resettlement, agitations). **Historic/heritage (s43–44):** community impacts (poor, low-income); impacts on culture/heritage; sites of archaeological value; visual resources; natural areas of scenic value; listed historic areas; other impacts (access to jobs, food, energy, aesthetics); examples Hazrat Ganj (Lucknow), Taj Mahal (Agra).
- **World Bank impact table for a town project (s46):** Social environment – traffic, development of social economy; land use; demolition and resettlement, traffic, irrigation works. Living environment – traffic, exchange, employment, income and safety. Ecological – water and soil loss. Water – water pollution (construction/operation), accident risk. Acoustic – traffic noise. Air quality – raised dust, TSP; harmful substances in tail gas (NOx, CO).
- **Approaches (s47–48): World Bank** – findings of country-specific environmental studies; national environmental/climate action plans, policy framework, legislation, institutional capabilities; obligations under international treaties. India: **NAPCC** missions (National Solar Mission, National Water Mission, Green India Mission). **EU** – compare baseline and alternatives; assess impacts qualitatively and quantitatively; group alternatives with same impacts and highlight differences. EU = political & economic union of **27** European countries with single currency.
- **Conclusion (s49):** EIA = multi-stage process identifying externalities, especially negative impacts, suggesting mitigation; screening, scoping, baseline collect data to predict impacts.

Important Distinctions to Remember

- **Screening** (is EIA needed?) vs **Scoping** (boundaries; TOR) vs **Baseline** (current status) vs **Prediction** (forecast).
- Standardized (inclusion/exclusion lists) vs Customized (case-by-case categories A/B/C).
- World Bank A (significant; type/location/sensitivity/scale) / B (potential adverse on populations or important areas) / C (minimal; no further EA).
- Net project area (right of way) < total project area (up to buildings) < area of direct influence.

Image slides read: s3, 4, 5, 6, 8, 9, 10, 12, 22, 23, 24, 30, 32, 45(blank), 46. s45 shows only the title (content absent in PDF).

Lecture 13 — EIA Processes-II (43 slides)

Mitigation L13·s3–16

- Stages: previous lecture = project proposal…impact assessment; **this lecture = Mitigation, Impact Statement, Review, Follow-up**.
- **Mitigation process:** impact assessment reveals possible damaging effects; mitigation = **measures to reduce or remove adverse environmental impacts**. Example: roadside plantations on highways reduce noise, limit dust, increase green cover — **National Green Highway Mission**.
- **Mitigation measures are:** (1) preventive measures that avoid impacts; (2) measures limiting **severity and duration**; (3) **compensation mechanisms** for unavoidable impacts. Example: **Project Tiger** (Govt. of India).
- **Objectives (s6):** identify actions that most reduce impacts and are practical/feasible; reduce severity to lowest possible level over project lifetime and avoid **irreversible and cumulative** effects; include activities to **monitor compliance**; identify risks of accidents, malfunctions, emergencies in all phases.
- **Project Tiger (s7–8):** rapid decline of tigers due to urbanization/development; **launched 1973**; positive results later. Tiger counts 2010→2014: Sunderbans 70→76; Northern Bengal —→3; Uttarakhand 227→340; Uttar Pradesh 118→117; Bihar 8→28; Andhra Pradesh 72→68; Chhattisgarh 26→46; Madhya Pradesh 257→308; Maharashtra 169→190; Odisha 32→28; Rajasthan 36→45; Jharkhand 10→3; Karnataka 300→406; Kerala 71→136; Tamil Nadu 163→229; Goa —→5; Assam 143→167; Arunachal Pradesh —→28; Mizoram 5→3.
- **Mitigation hierarchy (s9, BBOP 2009):** predicted impacts → Step 1 **Avoid** → Step 2 **Mitigate** → Step 3 **Restore** → Step 4 **Offset** → **No net loss**; further **Contribution** = **Net gain**.
- **Examples:** National Green Highway Mission (greening of NH corridors with farmers, private sector, Forest dept.) (s10); replanting vegetation on slopes; trees to limit noise/dust around housing (s11); **noise barriers along rail lines** – especially if rail passes through populated area; **Delhi metro has used it** (s12); wildlife passages – help animals navigate, maintain natural drainage, help local population access nearby locations (s13); **bamboo/rope bridge for reptiles** (snakes, lizards) in Uttarakhand (s14); **smog towers** (Delhi) to reduce pollutants in ambient air (s15); flora conservation – **Western Ghats** most diverse tropical region (s16).

Impact management; R&R; contingency L13·s17–24

- **Impact Management Plan (IMP):** series of plans and protocols to **manage and monitor** identified mitigation measures and risks. Implementation depends on: type of project; identified impacts; risks associated; natural and social context.
- **Elements (s19): Mitigation measures** (summary, linked to impact analyses) → **Monitoring** (plans for monitoring impacts and effectiveness of mitigation) → **Capacity development** (capacity-building needs to involve stakeholders) → **Implementation schedule and cost estimates** → **Integration** (EMP fits with planned project and mitigation actions).
- **Resettlement & Rehabilitation (R&R) (s20–21):** minimize number of people affected/displaced; collaborate with local authorities; community outreach; prepare for new jobs if livelihood changes; prepare for living in new region; overall **similar or better lifestyle**.
- **Contingency plan (s22):** clearly establishes lines of action, responsibilities, functions and quick effective responses to emergencies; includes assessment of possible risks and external contingencies (natural events, disasters) and potential threats to population health and ecosystem structure. Examples: industrial accidents/fires protocols; flood, earthquake safety.
- **Bhopal Gas Tragedy (s24): 1984**; highly toxic **Methyl Isocyanate (MIC)** gas leaked from pesticide plant (Union Carbide); **3787 died**, **574336** severely affected; proper contingency plan could have saved many lives.

EIS, review, public participation L13·s25–35

- **EIS** (defined by International Chamber of Commerce as a management tool of systematic, documented, periodic and objective evaluation…) — **final stage of an EIA**. Ideal contents: executive summary; policy, legal and administrative framework; description of environment; detailed description of project; significant environmental impacts; identification and analysis of alternatives; mitigation action/management plan; environmental management plan; monitoring program; knowledge gaps; public involvement; list of references; appendices (reference documents, photographs, unpublished data); Terms of Reference; consulting team composition; notes of public consultation sessions.

- EIS report formats shown for **IDB (Inter-American Development Bank), EU, World Bank** (s26–28): image text not preserved (blank in PDF).
- **EIS review (s29):** by **competent authority** as per legal requirements; in India by the government (**MoEF**). s30 "Objectives of EIS Review" lists 1–4 but item text is absent from PDF.
- **Public participation (s31–34):** goal = engage key stakeholders (citizens, NGOs, agencies, authorities, interest groups) to input on planned development, especially impacts affecting livelihoods. In practice consultation is done **mainly during review stage** (s32). **Features of effective participation (s33, Aschemann 2004):** inform and involve interested/affected stakeholders; explicitly address comments, concerns and inputs (in documentation and decision making); ensure all relevant actors/parties involved or represented; safeguard sufficient access to all relevant documents and information. s34 steps (World Bank 2010): image text absent.
- **Licensing (s35):** the **outcome of the review process**.

Follow-up / monitoring L13·s36–40

- Follow-up/Monitoring ensures: **accuracy of original predictions**; **degree of deviation**; possible reasons for deviation; whether mitigation achieved its objectives. Information contributes to improvement of future EIA (more accurate predictions).
- Monitoring process shown for World Bank (s37, text absent), **Asian Development Bank** (international organization which funds development projects across Asian countries, s38), **EBRD – European Bank for Reconstruction & Development** (s39).
- **General shortcomings of EIA reports (s40, Netherlands EIA Commission):** description of proposal does not cover key features (e.g. omits pipeline/facilities); key problems affected by the proposal not described; sensitive elements in affected environment overlooked; alternatives do not comply with environmental regulations/standards; appropriate mitigating measures not considered; serious environmental impacts or risks not described or incorrectly described; insufficient or outdated prediction models used.
- **Conclusion (s41):** EIA balances development and environment; without EIA harm to environment and indigenous population; proper implementation ensures sustainable growth and community benefits.

Important Distinctions to Remember

- Mitigation hierarchy order: **Avoid** → **Mitigate** → **Restore** → **Offset** (no net loss) → Contribution (net gain).
- **IMP** (manage/monitor mitigation) vs **contingency plan** (emergencies) vs **R&R** (displaced people).
- **EIS** = final stage of EIA; **review** by competent authority (MoEF in India); **licensing** = outcome of review; **public consultation** mainly at review.
- Project Tiger 1973; Bhopal 1984 (3787 died; 574336 affected).

Image slides read: s3, 8, 9, 19, 40. **Content absent in the PDF:** s26, 27, 28 (EIS formats), s30 (objectives), s34 (steps), s37 (World Bank monitoring), s38/s39 diagrams.

Lecture 14 — Methodologies of EIA (45 slides)

List of methods L14·s2–3

8 methods: Ad hoc · Checklist · Matrix · Network · Overlay · Environmental index (factor analysis) · Cost/Benefit analysis · Simulation.

Ad-hoc method L14·s4–10

- **Approach (s4):** indicate broad areas of possible impacts → list composite environmental parameters likely to be affected → **team of specialists** who identify impacts in their area of expertise → nature of impacts considered.
- **Features: intuitive and qualitative**; generally used in **screening process**.
- Common criteria (s6): air quality, noise, health and safety, groundwater, natural drainage, endangered species, wildlife, natural vegetation, economic values, exotic vegetation, social characteristics, recreation, visual description, grazing, public facilities, open space, services.
- Common practice: **committee of experts decides by consensus or majority**. **USA:** team of specialists with minimal guidance beyond NEPA requirements. **India:** similar process during screening to decide whether to do impact assessment.
- **Alternative reservoir example (s9):** 4 alternatives × parameters (air quality, noise, wildlife, local population, water quality); **rank 1/2/3/4** each alternative on every parameter and assess best option.
- **Pros/cons (s10):** Pro – simple, can be performed without training. Cons – no relative weighting or cause-effect relationship; suggests only broad areas; no assurance all relevant impacts studied; lacks consistency; inefficient (effort to assemble a panel each time).

Checklist method L14·s11–21

- Mainly used in **Screening & Scoping** stages. "Checklists are strong indicators of impact identification; effectively garner attention and awareness of audience."

Type	Features
Simple (s13–14)	List of parameters, **without guidelines** on interpretation; further needs measurement/data/prediction. Example (World Bank 2011, township project): factors × construction/operating period. Symbols: **Solid star = obvious effects; Star = usual effects; Circle = slight effects**
Descriptive (s15–16)	List of environmental factors + information on measurement, impact prediction and assessment. Example: **Air quality** – current condition (PM10, PM2.5, NOx, SOx), area affected, no. of people affected; **Water quality** – current status of water bodies, dissolved oxygen level, possible pollutants, effects on population; **Noise** – change in noise level, frequency of occurrence
Scaling/Weighting (s17–20)	List of factors + importance; weight per importance; needs experts for scaling. Example weights (total 100): **Air quality 40, Noise 15, Wildlife 10, Local population 20, Water quality 15**. **Limitations:** danger of equal importance to every impact; numerical values subjective; no provision for dynamic probabilistic trends, mitigation, enhancement, monitoring; cannot identify higher-order effects, impacts, interactions

Matrix method L14·s22–24

- Example (s23): rows = environmental components (**flora, fauna, air quality, water quality, population density, employment, traffic**); columns = project action – **Construction** (utilities; buildings residential + commercial) and **Operations** (residential, commercial, parks & open space). Pros/cons text (s24) not present in PDF.

Network method L14·s25–28

- Example 1/2 (s26, Esraz 2017): possible adverse impacts as chains **initial condition → consequent condition → effects**, e.g. increased surface runoff (E) → flooding (H) → gullying and erosion (K); pollution of ground water (F) → degradation of water supply (I) → health hazard (L); removal of topsoil (G) → decreased fertility (J) → death of flora (M). Corrective actions (e.g. place intermittent flower beds; plant shrubs) and control mechanisms (e.g. building code; plant shrubs). Branches: A→F→I→L (1); B→E→H→K (2); D/C→G→J→M (3).
- Example 2/2 (s27): impact scores on 1–10 interval scale (magnitude, importance) and probability of occurrence. Branch 1 = (2)(5)(.5)+(2)(9)(.6)+(2)(10)(.9) = **33.8**; Branch 2 = (5)(3)(.8)+(5)(3)(.7)+(4)(5)(.7)+(3)(7)(.7) = **51.20**;

Branch 3 = (3)(4)(.3)+(3)(4)(.4)+(2)(5)(.8)+(1)(6)(.8) = **21.2**; **Grand Network Index = 33.8+51.2+21.2 = 106.2**. (Pros/cons s28 absent.)

Overlay method L14·s29–32

- Set of maps of project area's environmental characteristics (physical, social, ecological, aesthetic); **separate mapping of critical features at the same scale as project site plan** (wetlands, steep slopes, soils, floodplains, bedrock outcrops, wildlife habitats, vegetative communities, cultural resources).
- **Old technique:** features mapped on **transparent plastic** in different colors. **Modern: GIS** with skilled personnel; aspects as layers with color coding (Zhang et al. 2011). (Pros/cons s30 absent.)

Simulation method L14·s33–34

- **Micro method** for a very specific situation; **experimental**; scope limited to a **single activity**; assesses impact of any product or small-scale service. Example: electric vs fossil-fuel vehicles compared in standard conditions for exact emissions and cost per km.

Cost-benefit analysis L14·s35–38

- **Social Cost-Benefit Analysis (SCBA):** assessing desirability of projects in **public as opposed to private sector**; identification of costs and benefits (social, environmental, economic); measurement in **monetary terms**; effect of risk and uncertainty during project life cycle; generally used by international institutions such as World Bank.
- **Stages of SCBA (s37):** 1 financial profitability based on market prices → 2 shadow prices for resources to arrive at net benefit at economic process → 3 adjustment of net benefit for projects impact → 4 adjustment for impact on income distribution → 5 adjustment for goods whose social values differ from economic values.
- **Limitations (s38):** measurement of social costs/benefits formidable; intangible, monetary evaluation subjective; comparison with future projects difficult; impacts differ from case to case.

Environment index method; EPI L14·s39–42

- **Standards predefined**; ranking for various parameters against set standards; **alternatives ranked by overall score**. Based on this, **EPI (Environment Performance Index)** ranks countries.
- **EPI:** international ranking mechanism assessing countries' efforts against climate change; uses **distance-to-target technique** (best and worst performance = scores 100 and 0); **32 indicators into 11 issue categories and two policy objectives**, with weights as % of total score.
- **Weights (s41): Environmental Health 40%** (Air quality 20%; Sanitation & drinking water 16%; Heavy metals 2%; Waste management 2%); **Ecosystem Vitality 60%** (Climate change 24%; Biodiversity & habitat 15%; Ecosystem services 6%; Fisheries 6%; Water resources 3%; Agriculture 3%; Pollution emissions 3%). Indicators shown include PM2.5 exposure 11%, household solid fuels 8%, sanitation 6.4%, drinking water 9.6%, CO_2 13.2%, CH_4 3.6%.
- EPI ranking map (s42) bands: 1–36, 37–72, 73–108, 109–144, 145–180, NA.
- **Conclusion (s43):** methodologies chosen by project requirements/conditions, time constraints, data and resources; each has benefits and limitations.

Important Distinctions to Remember

- **Ad hoc** (intuitive, qualitative, expert team; screening) vs **Checklist** (screening & scoping; simple/descriptive/scaling) vs **Matrix** (components × actions) vs **Network** (impact chains; Grand Network Index) vs **Overlay** (maps; GIS) vs **Simulation** (micro, single activity) vs **CBA** (monetary) vs **Index/EPI** (predefined standards, ranks).
- Simple (no guidelines) vs Descriptive (measurement/prediction info) vs Scaling (weights).
- EPI: 32 indicators, 11 categories, 2 objectives; Environmental Health 40% / Ecosystem Vitality 60%.

Image slides read: s3, 4, 7, 8, 9, 11, 12, 14, 19, 23, 26, 27, 36, 37, 40, 41, 42. **Content absent in the PDF (blank slides):** s18 (key features of scaling checklist), s21 (checklist pros/cons), s24 (matrix pros/cons), s28 (network pros/cons), s30 (overlay pros/cons). s10 (ad-hoc pros/cons) was extracted from text. s34 simulation example photo only.

Lecture 15 — EIA Process in India (49 slides)

Policies and acts L15·s3–8

Policy / Act	Key points
Environment (Protection) Act, 1986	Central Government has power to take all measures it deems necessary to protect/improve environment quality and prevent/control/abate pollution; **legal base for EIA process in India**
National Forest Policy, 1988	Ensure environmental stability and ecological balance; target: **one third area under tree/plantation cover**. Environment protection in **Articles 48A and 51A(g)** of Constitution; strengthened by judicial interpretation of **Article 21** (protection of life & personal liberty)
National Conservation Strategy & Policy, 1992	Sustainable and equitable use of resources; conserve natural and man-made heritage; prevent/control deterioration of land, water, air; minimize adverse consequences of development projects; restore degraded areas
Policy Statement on Abatement of Pollution, 1992	Pollution **prevention** instead of conventional end-of-the-pipe treatment; best available and practicable technologies; clean and low-waste technologies, waste minimization, reuse/recycling, water quality improvement, environment audit, natural resource accounting
National Water Policy, 2012	Take cognizance of existing situation; propose framework for system of laws and institutions; plan of action with unified national perspective
National Environmental Policy (NEP), 2006	Better livelihoods through resource conservation; greater role to local bodies; **EIA = principal methodology for appraisal and review of new projects**; coordination among stakeholders; protection integral to development

EIA Notification 2006 L15·s9–29

- In accordance with National Environment Policy objectives; legally based on EP Act 1986; restrictions/prohibitions on new projects and modernization of existing ones; applicable **everywhere in India including territorial waters**.
- **SEIAA (s10):** constituted by Central Government; **three members** including Chairman & Member-Secretary; State/UT forwards names; professionals/experts; decisions **unanimous** and taken in a meeting otherwise majority prevails; **constituted after every 3 years**.
- **Environment Clearance (EC) (s11–12):** legal requirement permitting project to start; needed before any construction or land preparation (except securing land). Needed for: all new projects/activities listed in 2006 notification; expansion beyond specified capacity limits; change in product-mix beyond specified range.
- **Process (s13):** categorization through screening → **A category: MoEF&CC-EAC** (Environment Assessment Committee); **B category: SEIAA**. **2006 has only 2 categories: A = high impact, B = medium impact (s14).**
- **Transport categories 2006 (s15–16):**

Project	Category A	Category B
Airport	All airports	—
Oil & gas pipeline transport	All oil & gas transport projects	—
Aerial ropeway	—	All aerial ropeway projects
Harbor/Port	> 5 million TPA cargo handling (excl. fishing harbors)	< 5 million TPA
Highway	New national highways and expansion of National highways > 30 km, additional right of way > 20 m involving land acquisition and passing through more than one State	New state highways and expansion of National/State highways > 30 km with additional ROW > 20 m involving land acquisition

  TPA = tonnes per annum. **Railway projects do not come under EIA notification 2006** but an EIA report is needed for other clearances such as forest clearance.
- **Change of category (s17):** Category B treated as A if located in whole or part within **10 km** of boundary of: protected areas (Wildlife (Protection) Act 1972); notified eco-sensitive areas; critically polluted areas (CPCB); inter-State and international boundaries.
- **Application (s18–21):** prescribed **Form 1/Form 1A** + pre-feasibility project report (conceptual plan where not required). **Form 1A** = check-list form, Yes/No, **93 parameters in 10 categories** of impact (e.g. vegetation, air

quality, water quality). **Pre-feasibility report:** project background; description; site analysis; proposed infrastructure; R&R plan; schedule & cost estimates; overall analysis (social, economic, environmental). **Conceptual plan:** project description; site location; important features; impacts (if any).

- **Scoping (s22–23):** comprehensive ToR by applicant; **site visit by sub-group of appraisal committee**; ToR conveyed **within 60 days**; if not conveyed within 60 days, provided ToR **assumed final**; approved ToR displayed on MoEF&CC website; EC may be rejected on recommendation of appraisal committee; reasons communicated within 60 days.
- **Public consultation (s24–26):** public hearing (on-site) and written responses; conducted by **SPCB / UTPCC**; if absent, independent agency assigned by EAC/SEAC; if not possible, appraisal committee notes in report and proceeds. **Bhatta-Parsaul case (s25):** 2011, U.P. govt acquired land for **Yamuna expressway** near Greater Noida → uproar, violence, political & social crisis. **Exceptions (s26):** expansion of roads/highways **not involving further land acquisition**; national defense/security/strategic projects; specified projects within existing industrial estates & irrigation projects.
- **Appraisal (s27):** grant EC on stipulated terms or reject with reasons; based on Form 1/1A and info after site visit; appraisal committee recommendations placed before MoEF&CC for decision **within next 15 days**; appraisal completed **within 60 days** of receipt of final EIA report.
- **Post-EC monitoring (s28): half-yearly compliance reports** (hard & soft copy) on **1 June and 1 December**; reports are **public documents**.
- **Transferability (s29):** EC may be transferred during validity to another legal person with **written "no objection" from transferor**; same terms and validity period; no reference to EAC/SEAC necessary.

EIA Draft Notification 2020 L15·s30–46

- Central govt notified new draft in **March 2020** for public comments; comes into force on date of publication of final notification in Official Gazette; legally based on EP Act 1986.

Aspect	EIA notification 2006	EIA draft 2020
Categories	2 (A/B)	3 (A / B1 / B2)
Terms	Various terms not defined	Terms are defined
Data	Yearly data of impacts required	Data of **one season** sufficient
Monitoring frequency	Every 6 months	Yearly
Monitoring report	By third party [as written]	Submitted by project proponent

- **B1/B2 (s32):** under 2006 some Category B projects appraised at central level (treated as A); 2020 splits B into B1 and B2.

Transport categories in draft 2020 (s33–35)

Project	A	B1	B2
Airports and heliports (terrestrial & water)	All airports	—	Water aerodromes for commercial use; heliports for commercial use
Oil & gas pipeline transport	All oil & gas transport projects	—	—
Elevated roads / standalone flyovers / bridges	—	—	> 1,50,000 sq. m built-up area
Aerial ropeway	—	—	All projects in notified ecologically sensitive areas
Harbor/Port	> 5 million TPA cargo (excl. fishing harbors)	< 5 million TPA cargo; > 30,000 t/yr fish handling	All projects in respect of inland waterways

Highway / Expressways / Multi-modal corridors / Ring roads	New National Highways/Expressways/corridors/Ring Roads; expansion or widening of existing NH/Expressways/etc. by length > 100 km with widening or ROW > 70 m on existing alignments, realignments or bypasses	All new State Highway projects; State Highway expansion projects in hilly terrain (above 1,000 m above MSL)	Expansion/widening of existing NH/Expressways/etc. by length between 25 km and 100 km with widening or ROW > 70 m; expansion/widening of existing State Highways (500 m to 1000 m above MSL)

- **New definitions in 2020 (not defined in 2006), s36–39: ACO** – accredited EIA consultant organization accredited by **NABET of Quality Council of India (QCI)** or any agency notified by Ministry. **Capital dredging** – one-time removal of virgin material from sea-bed to create/deepen a shipping channel for larger ships (inside/outside ports/harbors and channels). **Border area** – within **100 km aerial distance from Line of Actual Control** with bordering countries. **Certificate of green buildings** – under rating programmes **GRIHA, IGBC, LEED, EDGE**. **District/Divisional Level Expert Appraisal Committee** – constituted at district/divisional level on State/UT recommendation.
- **B1 exceptions (s40):** a B1 project is appraised at **Central level** if located within **5/10 km** (varies by project type) of protected areas, critically polluted area, eco-sensitive area, inter-State/UT boundary; or within boundary of severely polluted area or eco-sensitive zone.
- **Reduced monitoring (s41):** earlier every six months → now **once a year**; compliance report filed by proponents (possible discrepancies). Examples of weak monitoring: **Vizag gas leak** (polymer plant, styrene gas, May 7; running 20+ years without clearances; **11 died, ~1000 sick**) and **Assam refinery** (Tinsukia district, natural gas blowout and fire May 2020; operating 15+ years without prior clearances).
- **Seasonal data (s44–46):** draft permits collecting data during **one season** (**except river valley projects**); data change by season → may hamper baseline; examples: summer air pollution and Himalayan river flow changes.
- **Conclusion (s47):** policies/acts conserve at various stages; EIA important; draft 2020 raises concerns among some experts.

Important Distinctions to Remember

- 2006: **A (MoEF&CC-EAC)** and **B (SEIAA)** vs 2020: A / B1 / B2.
- 2006 yearly data & 6-monthly third-party monitoring vs 2020 one-season data & yearly proponent-submitted monitoring.
- Time limits: ToR within **60 days**; recommendations to MoEF&CC within **15 days**; appraisal within **60 days**; compliance reports **1 June / 1 Dec**. SEIAC: 3 members, reconstituted every 3 years.
- Form 1A: 93 parameters, 10 categories. Highway A (>30 km, ROW >20 m, more than one State) 2006 vs 2020 (>100 km, ROW >70 m).

Image slides read: s13, 15, 16, 33, 34, 35. The 2006 table text layer misaligned the aerial-ropeway row; the rendered slide (used here) shows ropeways in Category B. s31 text used as extracted.

Lecture 16 — Global practices in EIA process (40 slides)

SEA, screening, cumulative impacts L16·s3–16

- **SEA legal provisions (s3):** integrated within EIA legislation – **Kenya, Tanzania**; separate laws – **Indonesia, Canada**; integrated in other sectoral legislations (transport, energy) – **New Zealand**.
- **Parameter selection approaches (s4): Environment-centered** (listing factors/criteria for likely impact); **Development-centered** (listing development activities); **Hybrid** (between the two). **India follows hybrid approach** for screening (checklist has both) (s5–6): examples – emissions from fossil fuel combustion, production processes, materials handling, construction, dust/odours (environment-centred); land-use change, clearance of land/vegetation/buildings, new land uses, pre-construction investigations, construction works (development-centred). **EU** checklist (s7) is also hybrid (physical changes, natural resource use, harmful substances, solid waste, release of pollutants).
- **EU (s8):** member countries classify projects into **two categories (I & II)**; case-by-case examination, threshold criteria, or hybrid.
- **Peru (s9):** three categories – **I**: only EIS (low impacts); **II**: Semi-detailed EIA (EIA-SD; moderate impacts); **III**: Detailed EIA (EIA-D; significant impacts) (Law 27446, Art. 4).
- **Kenya (s10):** one of the few requiring an **environmental practitioner at screening stage**; knowingly giving false/misleading information is an offence (law text: imprisonment up to 24 months or fine not more than two million shillings or both).
- **Cumulative impacts at scoping (s11–12):** regional perspective rather than local. **China** – no mention of cumulative impact assessment; **Canada** – study of cumulative effects of physical activities in a region (Canadian EIA legislation 2012). Example: **Delhi** winter pollution rise partly from **stubble burning in neighboring states** → cumulative impacts need regional-level management.
- **Mitigation for individual impacts (s13–14):** China – mitigation considered during scoping; pollution-control installations must be designed, built, commissioned (Environment Protection Act 2014, Arts 40-42). Examples: noise barriers; treating sludge before dumping in rivers; efficient machines.
- **Offsets (s15–16): Brazil** – projects subject to licensing must offset impacts on protected areas. **Mexico** – Sustainable Forest Development Act **2003** requires offsets for land-use change in forests; Environmental Liability Act **2013** requires offsets only when impacts not predicted/approved in EIA and are an environmental offence. Example **Antamina mines (Peru): Polylepis** shrub habitat indirectly impacted; EIA: overall impacts limited, key concern = destruction of endangered Polylepis; large-scale conservation programme carried out.
- **Responsibility for EIA reports (s17):** government agency – **USA**; licensed/registered consultants – Vanuatu, Cameroon, Peru, China, Mongolia, Oman, Lebanon; individual with certification – **Indonesia**. India: agencies registered under **QCI** or notified by MoEF&CC.

Public participation; review; follow-up; cultural impacts L16·s18–27

- **Canada (s18):** public invited to comment for **20 days** after project subject to screening and at other stages; comments must be considered by Canadian Environmental Assessment Agency; **multi-stage, from screening to monitoring** (against general practice elsewhere).
- **Definition of public: China** (2014 Act) – full EIA text public; hearings/submissions restricted to residents, environmental groups and others at the project site likely affected. **Denmark** – natural/legal persons directly or indirectly affected; associations/organizations with protection of landscape, cultural heritage, natural/environmental interests as objective.
- **Baltimore light rail (s21):** 52 pre-scoping "Listening Tours"; initial scoping meetings/written comments; published reports; community liaisons; **four public hearings**; Citizens' Advisory Council; Speaker's Bureau; resource hubs.
- **EIA review (s22–23):** EU guideline – review EIS before decision, by govt agencies or professionals on behalf of govt. **Cameroon** – ESIA report to Ministry of Environment, Nature Protection and Sustainable Development; on-site visits to verify; outcomes to **Inter-ministerial Committee**.
- **UNEP best features for follow-up/monitoring (s24):** sufficient information at regular intervals; manage adverse impacts using monitoring data; ensure indigenous community participation; expand powers of competent enforcement agencies; create regional monitoring networks.
- **Cultural impacts: Canada (s25)** – for aboriginal peoples: effects on health and socio-economic conditions; physical and cultural heritage; current use of lands/resources for traditional purposes; structure/site/thing of

historical, archaeological, paleontological or architectural significance. **Laurentia Project (s26)** – Port of Quebec deep-water wharf, Beauport: EIA found significant adverse effects (fish and habitat, air quality and human health, socio-economic conditions, traditional use by Indigenous peoples) → **project not commissioned**. **New Zealand (s27)** – effects on aboriginal/indigenous tribal peoples: health & socio-economic; physical & cultural heritage; traditional land use; historical/archaeological/paleontological/architectural sites; impact on community knowledge and aboriginal traditional knowledge.

Comparative overview: developed vs developing vs India L16·s28–32

Aspect	Developed countries	Developing countries	India
Legal basis	Well-framed legislation (Canada: Canadian Environmental Assessment Act; EU: Directive on EIA 1985)	Lack of formal legislation; EIA not mandatory in many African countries	Formal; enacted by amendment in Environment Protection Act 1986
Public participation	Active involvement of all participants at early stages	Limited involvement of public/government agencies in initial phases → poor representation, lower report quality	Limited, in initial phases
Impact type & alternatives	Integrated approach incl. social and health; proper consideration of alternatives	Mainly environmental; poor on social/health; alternatives absent	No provision to cover landscape and visual impacts; alternatives same as developing
Language	Reports in local language	One primary language, not local (executive summary sometimes translated)	Mostly in English, not local language
Technical approach	Multi-disciplinary, experts	Lack of trained professionals → inadequate/irrelevant reports	Done by consultants; selection criterion is **fees/cost** rather than expertise
Expertise	Many individuals able to design/conduct/review EIAs; teaching also in industrial countries	Developing slowly; students go to developed countries	Developing

Gaps/loopholes L16·s33–37

- **Project splitting / salami slicing:** separate project into small parts each below thresholds to bypass EIA. Examples: **Spain** – Valencia–Tarragona railway line (Las Palmas–Oropesa section), part of **Mediterranean corridor**; Spain forced to carry out EIA. **UK** – White City and Crystal Palace urban development projects divided; UK failed obligations; project **stayed by European court**.
- **Measures (s36): Portugal** – apply EIA below thresholds if necessary; **Sweden** – assess scope of project in screening phase; **Denmark, Finland, Spain** – case-by-case screening; **Germany, UK** – regulation of requirements to consider a project independent; **Austria** – consider all associated developments part of a project.
- **Exemptions (s37): Nigeria** – exempt if in interest of public health/safety; **Indonesia** – exempt if location already had an EIA or within district/city with detailed spatial plan.
- **Conclusion (s38):** approaches vary by country; adopt best practices; more constructive public participation; "implemented in spirit, not just words".

Important Distinctions to Remember

- Parameter approach: environment-centered / development-centered / **hybrid (India, EU)**.
- Peru 3 categories (EIS / EIA-SD / EIA-D) vs EU 2 categories (I & II) vs India 2006 (A/B).
- Canada = 20-day comments, multi-stage participation, cumulative effects; China = no cumulative impact mention.
- Kenya = practitioner at screening; Indonesia = certified individual prepares EIA; Nigeria/Indonesia exemptions.
- Splitting examples: Spain (Mediterranean corridor), UK (White City, Crystal Palace).

Image slides read: s5, 6, 7, 9, 10, 25, 36. s9/s10 contain legal-text images (Peru Law 27446; Kenya Cap. 387) used as shown.

Lecture 17 — EIA Case Study-I: Mumbai–Ahmedabad High Speed Rail (MAHSR) (39 slides)

Project L17·s3–11

- First proposed in Indian Railways' **"VISION 2020"** (Ministry of Railways); high speed corridors allow up to **350 kmph**; to use **Shinkansen** trains from Japan; connects Mumbai–Ahmedabad; completion expected **2028**.
- **Shinkansen (s4):** popularly "Bullet Train"; famous for **zero casualty** from train accident in **50 years** with **10 billion passengers**; literal meaning "**New trunk/main line**".
- **EIA report (s5):** proponent = **National High Speed Rail Corporation Limited (NHSRCL)**. Prepared by: **GPS Technologies Pvt. Ltd** – S-EIA report, primary/secondary environmental data; **NHSRCL / RITES** – land acquisition plan; **ARCADIS** – Resettlement and Rehabilitation Action Plan. (RITES = Rail India Technical and Economic Services.)
- **NHSRCL (s6–7):** incorporated **12 February 2016** under Companies Act 2013; to finance, construct, maintain, manage HSR corridor; modelled as **Special Purpose Vehicle (SPV)** in joint sector; equity by Central Government (Ministry of Railways) and governments of **Gujarat and Maharashtra**. SPV = joint venture/legal entity for a specific objective; collaboration of multiple parties; reduces risk for any single entity.
- **Location (s8): 508 km**; **12 stations** (Mumbai (BKC), Thane, Virar, Boisar, Vapi, Billimora, Surat, Bharuch, Vadodara, Anand, Ahmedabad, Sabarmati); Gujarat 1,100 hectares/195 villages; Maharashtra 350 hectares/104 villages.
- **Benefits (s9):** travel time **6.5 hrs → 2.5 hrs**; connectivity boost to Palghar, Valsad, Vapi, Bharuch; technological enhancement; jobs – construction **~20,000**, operations & maintenance **4,000**, **16,000** indirect; decongestion of rail, road, air; GHG reduction; state-of-the-art manufacturing and software ecosystem.

Aspect (s10)	MAHSR	Railway/Road/Air
Land use	**2.8 ha/km**; minimal change in pattern	**9.3 ha/km**; change at large scale
Travel time	**2 hr**; better connected to local transport	6–12 hr (rail), 1.25 hr (air); airports not easily accessible
GHG	10% annual reduction wrt no change; **81400 tons CO_2 over 15 years**	[slide: "6 times by 20250"; 1.4 metric tons CO_2/km by 2050]
Fuel consumption	Significant reduction due to mode change	Steep increase if no change

- **Financing (s11):** estimated cost **INR 1.08 lakh crores**; Government of Japan **soft loan of 12 billion USD at 0.1% interest**; repaid over **50 years** including **15-year grace period** (period when annual payments not required); through **JICA** (Japan International Cooperation Agency).

Alternatives; regulations L17·s12–19

- **3 alternatives (s12):** Alt-1 HSR with viaduct/bridges and tunnels; Alt-2 HSR with embankment and bridges and tunnels; Alt-3 **No-HSR – Business-as-Usual (BAU)** (= "no change" situation).

Aspect (s14)	Alt-1 (viaduct)	Alt-2 (embankment)	BAU
Length	508.17 km	508.17 km	538 km (road)
Travel time	2 hr 07 min	2 hr 07 min	8 hr 52 min (car)
Travel demand by 2023 (per day)	40 thousand	40 thousand	38 thousand
Safety level	High	High	Low
Cost	1080000 million INR	709151 million INR	N/A
Environment impact	Minor damage; Thane creek not affected	Same	N/A
Land use	2.8 ha/km	3.2 ha/km	7.5 ha/km (conventional rail)

- **JICA (s15):** project categorized **Category A** per JICA guidelines for environmental and social consideration (**April 2010**), similar to World Bank guidelines; JICA = international lender organization supported by Japan.
- **Clearances (s16):** diversion of forest land; mangroves and wildlife (protected) areas; tree felling; coastal regulation zone; adverse environment impacts; occupational health & safety and labour laws.
- **Authorities (s17):** MoEF&CC; CPCB/SPCB; NBWL & SBWL (wildlife boards); State Forest Departments; **NGT** (National Green Tribunal); NCZMA & SCZMA; **DTEPA** (Dahanu Taluka Environment Protection Authority – taluka-level authority created to protect natural habitat of tribal people in tribal district of Palghar).
- **Acts (s18):** Environment (Protection) Act 1986; Air Act 1981; Water Act 1974; Indian Forest Act 1927; Wildlife (Protection) Act 1972; Maharashtra Felling of Trees (Regulation) Act 1964; Maharashtra (Urban Areas) Protection and Preservation of Trees Act 1975; Wetlands (Conservation and Management) Rules 2010.
- **Environment attributes (s19):** physiography & terrain; drainage; soil; geology & seismology; landscape; land use; ground subsidence; climate; climate change; water use/demand; water quality; water body; air quality; noise; vibration; ecology; protected areas; sensitive location; CRZ; forest & mangroves.

Impacts and mitigation L17·s20–30

Impact	Mitigation
Landscape (s20): viaduct Mumbai–Ahmedabad; bridges across water bodies; permanent change near stations	Suitable (less impact) viaduct design; landscaping, minimize temporary visual impacts; additional tree planting; landscaping at stations
Deforestation (s21): forest diversion **138.3269 ha incl. 32.3902 ha mangrove**; mangrove removal **24.3981 ha**; **60000 trees** expected to be cut	Compensatory afforestation and mangrove planting per regulation; **balancing culverts** in coastal/mangrove areas to maintain tidal water exchange
Sea level rise (s22): impact on alignment and station structures near coast at BKC-Mumbai (global rise **3.6 mm/yr** in last decade)	Station near Mithi river designed for sea-level rise of **1 m over 100 years**
Energy & GHG (s23): electrical energy for operation/stations; indirect GHG	Shinkansen technology more efficient than conventional Indian railways; energy-star rated equipment/fixtures; offset by savings from modal shift from private road travel
Dust & emissions in construction (s24)	Vehicles with **PUC certificate**; water sprinkling on unpaved roads; covered material transport; enclosure/barricading
Noise in construction (s25)	Enclosures around stationary equipment; noise barriers at sensitive receptors; exhaust mufflers; **PPE** for workers
Vibration (s26): tunneling, TCFS impact	Noise abatement measures in tunneling; noise barriers
Safety/comfort of community (s27): piling vibration, blasting	Night-time piling avoided near residential areas; **blasting in daytime only** with advance intimation
Flora (s28)	Verify with forest officials and mark trees to be cut; compensatory plantation with native or pollution-tolerant species; compensation for private trees by fruit-yield/timber value; minimal clearing; compensatory mangrove plantation
Fauna (s29)	No construction yard in forest areas; heavy machinery away; immediate soil compaction for embankment; quick clean-up of oil spills; **crossing structures (tunnels for animal movement)** at forest patches
Water bodies/drainage (s30)	No dumping in natural drains; adequate cross-drainage (longitudinal and median drains); augment capacity of existing drainage; drainage at construction site/camp to avoid stagnant pools

Public participation; monitoring L17·s31–37

- **Public meetings at 10 places** (Mumbai/Thane, Palghar, Valsad/Dadra Nagar Haveli, Surat, Navsari, Bharuch, Vadodara, Anand, Kheda, Ahmedabad); key issues: **compensation and religious structures**.
- **Grievance redressal (s32–33):** project passes through protected areas/tribal regions → social acceptability issues with NGOs/activists, tribal-rights organizations, land-acquired locals; grievance redressal is part of public participation. Mechanism functions: record, categorize and prioritise grievances; field visits; fix timeframe; convey GRC decision in writing. District-level GRC members: Resident Additional Collector (RAC); CPM or Dy.CPM (NHSRCL); representatives of PAPs (Sarpanch etc.); Site Environment Officer.

- **Operation-phase monitoring (s34):** social impacts; crop & vegetation; air quality; noise & vibration; water quality; flora and fauna.
- **EMoP (s35):** mitigation plans and monitoring executed by **SEMU (Social & Environmental Management Unit)**, one in each State (Maharashtra, Gujarat), under Project Implementation Unit (PIU) of NHSRCL; contractors implement EMP; supervision consultant; third-party HSE auditor.
- **Inferences (s36):** timeline shifted to **2028** due to delays in land acquisition; passes through eco-sensitive coastal zone and tribal areas; strict monitoring needed. **Conclusion (s37):** binding regulatory EIA ensures impact assessment, mitigation, less adverse practices, monitoring, public participation.

Important Distinctions to Remember

- Alt-1 viaduct (2.8 ha/km; ₹1,080,000 m) vs Alt-2 embankment (3.2 ha/km; ₹709,151 m) vs BAU (no HSR; 538 km road; 8 h 52 min).
- JICA = lender (Category A); NHSRCL = proponent/SPV; GPS Technologies = S-EIA; ARCADIS = R&R; SEMU executes monitoring.
- Soft loan 12 bn USD, 0.1%, 50 yrs incl. 15-yr grace. Travel time 6.5 h → 2.5 h.

Image slides read: s8, 10, 13, 14, 33, 35. *s10: "6 times by 20250" and sign-off wording are as printed on the slide.*

Lecture 18 — EIA Case Study-II: Hunan Xiangjiang Inland Waterway, China (42 slides)

Project L18·s3–14

- **Project:** construction of a multi-purpose river barrage, improve navigational draft along a **50 km reach** of Xiangjiang River, generate hydropower. Lender: **Asian Development Bank (ADB)**. EIA report by Hunan Xiangjiang Navigation Construction & Development Co., Ltd. (**2011**).
- **ADB (s4):** envisions prosperous, inclusive, resilient, sustainable Asia and Pacific; loans, technical assistance, grants, equity investments; **68 member countries including India**.
- **Location (s5):** middle-to-lower reaches of Xiangjiang River, tributary of the **Yangtze**; **Tugutang Complex ~38 km** upstream from Hengyang and ~**50 km** downstream from Jinweizhou Complex.
- **Main components (s6):** Tugutang barrage to raise water levels and flood dangerous shoals along 50 km; hydropower station **90 MW**, normal annual energy **358.2 MkWh**; **two new river wharves**; works in inundated areas incl. resettlement of farmers.
- **Need (s7–8):** hilly terrain/narrow valley constrain road and rail; poor navigation; industrial output along river = **60%** of province; freight predicted **0.20 and 0.29 billion tons** by 2020 and 2030 from **20 million tons in 2010**; energy consumption of unit transport ≈ **65% of railways and 15% of highway**; CO_2 emissions from water transport < **half of rail** and **22% of highway**.
- **ADB EIA (s10):** Safeguard Policy Statement (**2009**), like World Bank; project screened and categorized **Category A** (potential significant adverse impacts).
- **Chinese laws (s11):** 1989 Environment protection law; 1997 Prevention & control of environmental noise pollution; 1998 Protection of wildlife; 1998 Flood control law; 2001 Evaluation of environmental effects; 2002 Water law; 2004 Fishery law; 2005 Prevention & control of pollution by solid waste; 2008 Prevention & control of water pollution; 2009 Prevention & control of atmospheric pollution.
- **EIA steps in China (s12):** environmental baseline data; project alternatives; recommendations for minimizing/mitigating/eliminating impacts; analysis of extent and distribution of adverse impacts and consequences; opinions of affected persons; environmental monitoring program.
- **No-project scenario (s13):** river not navigable during dry season between lower and upper sections; cargo loaded off ships onto road/rail between Jinweizhou and Hengyang (**88 km**). **IWT advantages (s14):** geography suitable; lower operating cost per ton-km than road and rail; no road congestion/rail capacity issues; relatively safe and secure.

Location alternatives for barrage L18·s15–19

Aspect	Upstream	Middle	Downstream
Engineering feasibility	Suitable geology; topography more suitable → **preferred**	Suitable geology	Suitable geology
Navigation	Straight river section; depth 1.0–1.6 m low-water → preferred	Slight meandering; 90° turning section; affected by Dayuandu reservoir (2–3.5 m)	Slightly meandering; 1 km downstream dam site is 90° turn; Dayuandu dead water depth 2–3.5 m
Investment cost	Comparable to Middle	Comparable to Upstream	Higher (higher barrage) → Upstream or Middle preferred
Hydropower	Lower than Middle & Downstream	Comparable to Downstream	Comparable to Middle → Middle or Downstream preferred
Master-plan compatibility	Compatible	Compatible	**Conflicts with already built Dyuandu complex; not viable**
Excavation (million m^3)	**1.8**	2.9	2.1
Water quality	Upstream of major wastewater outfalls and waterworks inlets (positive)	Urban and industrial wastewater outfalls within reservoir area	Outfalls within reservoir; Hengnan County Waterworks water intake within reservoir
Aquatic ecosystem	Barrage coincides with fish spawning areas	Same as upstream	Same as upstream

Dredging	Major dredging downstream	Minor	**No dredging** (positive)
Land (cultivated, mu; **1 mu = 6.07 acre**)	Inundates **4 thousand** (positive)	4½ thousand	5 thousand
Flood & drainage	No impact on Yunji town (positive)	Adjacent to Yunji town	Adjacent to development downstream of Yunji town
Resettlement	Relatively few adjacent residential areas (positive)	Potential conflict with Hengnan County industrial park and proposed Hengyang airport	Left bank adjacent to Chejiang town, densely populated

Upstream location was selected.

Impacts and mitigation L18·s20–27

- s20 scoping overview (flow chart): operation of new barrage → decreased water flow velocity (sedimentation upstream up/downstream down; heavy-metal sedimentation; plankton/fish change from lentic to lotic); new physical barrier (restricts fish movement); upstream water level rise (figure too small to read); more sustainable transport and power vs alternatives → potential effects: resettlement of displaced communities, farmland waterlogging, slope instability, lower CO_2 footprint.
- s21 impact matrix (HydroChina Zhongnan EIA): construction-period actions (land occupation, vegetation removal, stockpiles, dredging, machinery, materials transport, concrete batching plant, stone quarries) and operation (initial filling, reservoir impoundment, transportation, energy generation, vegetation recovery) vs resources; legend: long-term beneficial (□), short-term beneficial (Δ), long-term negative (■), short-term negative (▲).

Impact	Mitigation
Hydrology (s22): flow change from coffer dams; decreased flows downstream during initial impoundment; river → reservoir flow characteristics; extreme runoff; lower velocity near wharfs	Coffer dams temporary; **reservoir filling programmed for wet season**; hydrology modified to minimize flooding risk
Water quality (s23): sewage and wastewater discharge; oil/fuel spills; lower diffusion	Collect construction water/runoff in sedimentation ponds; **two bio-filter sewage treatment plants** (one each side); treat oily water from workshops; sedimentation ponds for aggregate washing
Groundwater (s24): levels expected to rise **0.5 to 2.1 m**	Flooding mitigated by removing soil from nearby hilltops to raise land
Fisheries (s25)	Fishing prohibited in spawning season; coordinate opening of sluice gates for migratory fish; **fish pass** alongside barrage; **fish hatchery** and long-term programme for artificial release of fish spawn
Air quality (s26): dust/emissions in construction; operation emissions (mostly SO_2, NO_2)	Dust & emissions control programme; air quality monitoring during construction vs standards
Displaced population (s27)	Compensation for permanent land loss (land fee, resettlement subsidy, standing crop); temporary land loss (crop + restoration fee); houses/structures (+ relocation allowance, crop losses); business/employment; public facilities

Public participation; grievance; monitoring L18·s28–40

- **EIA hearing forum:** held **August 4, 2010**; most representatives supported. **Group investigation: 300** questionnaires handed out, all completed (**100%** response); **95%** agree project conducive to regional economic development. **Web publicity:** on Hengnan County government website and Immigration Bureau of Changning City; posted at least **10 days**; comment: scientific comparison and verification needed to choose best programme. **Field publicity:** first round – public notices in affected region (basic info, preliminary impact analysis, how to give views); second round – same locations with updated info incl. mitigation measures.
- **Public concerns (s33):** land acquisition/resettlement and environment; fear of inadequate compensation; response: compensation strictly per national provisions.
- **Grievance redressal (s34–35): Old mechanism** – TRO (Township Resettlement Officer), CRO (City RO), HRO (Hunan Province RO), XNCD (State council for development), RE (Resident Engineer), Contractor, Court of Law –

complicated, lacked clarity. **New simplified mechanism** – affected people, contractor, local government, resident engineer around **Public Complaints Centre XNCD** – simple and clear.

- **Monitoring during construction (s36–37):** wastewater quality; river water quality up/downstream; air quality **quarterly at three sites** near villages; noise at sensitive locations around Tugutang Dam site **one day a month over 12 occasions**; terrestrial ecosystems; population health – **health checks on 10% of construction workforce** with quarterly reporting on infectious diseases; soil and water conservation in wet season.
- **Monitoring during operation (s38–39):** domestic sewage wastewater at offices/ports; river water quality at **four locations**; sediments (heavy metals in riverbed); aquatic ecology (hydrologic/hydrodynamic characteristics, plankton, benthos, fisheries); terrestrial ecosystems – **three campaigns** (one pre-construction, two after: spring and autumn); population health general survey; vegetation recovery monitoring.
- **Conclusion (s40):** important for interconnectivity; beneficial over other modes; impacts assessed/mitigated; holistic participation; monitoring in place.

Important Distinctions to Remember

- Upstream (selected) vs Middle vs Downstream: lowest excavation (1.8), lowest cultivated land flooded (4 thousand mu), no Yunji impact; downstream conflicts with Dyuandu complex (not viable) but needs no dredging.
- Old (TRO/CRO/HRO/XNCD/RE) vs new simplified grievance mechanism (Public Complaints Centre XNCD).
- ADB Category A; 90 MW; 358.2 MkWh; 50 km reach; 68 ADB members.
- Survey: 300 questionnaires, 100% response, 95% agree.

Image slides read: s15–21, 34, 35. s20 text partly small; The upstream water-level figure on the flowchart is too small to read.

Lecture 19 — EIA Case Study-III: NH-707 hilly road, Himachal Pradesh (38 slides)

Project L19·s3–14

- **Gumma to Fediz (NH-707)**; MoRTH development of single/intermediate lane NHs; **National Highways Interconnectivity Project (NHIIP)** aided by the **World Bank**; EIA report by "Consulting Engineers Associates, Panchkula".
- **Benefits (s4):** reduction of **Vehicle Operating Cost (VOC)** (less roughness, improved alignment); improved financial sustainability for road development and maintenance; increased vehicle speeds; reduced passenger time costs.
- **EIA steps (s5):** study of background info; screening report; **Survey Part-1 (baseline data collection)**; **Survey Part-2 (socio-economic)**; analysis of practical impacts; meeting with stakeholders; mitigation measures adoption; **Environment Management Plan**.
- **Physical features (s6–8):** hilly tract with undulating terrain; elevation **2500 m to 3500 m** above MSL; deep valleys, river terraces, steep escarpments, streams. Geology: alluvial deposits, sandstone, clay top zone, variable rocks. Seismology: **moderately earthquake prone**. **Landslides:** soil stability major issue; landslips, mud flows, flash floods; rises in rainy season. **8 major landslide points** (chainage km): 95.530–95.580; 95.400–95.430; 96.520–96.570; 98.800–98.850; 101.170–101.270; 101.370–101.470; 101.690–101.730; 102.630–102.690.
- **Policies (s9):** National Forest Policy 1988; National Water Policy 2012; National Environment Policy 2006; National Conservation Strategy 1992; National Policy on Resettlement and Rehabilitation for Project Affected Families 2007. **Laws (s10):** EP Act 1986; Forest (Conservation) Act 1980 (amended 2003 & 2004); Wildlife (Protection) Act (amended 2002); Air Act 1972 [as written]; Water Act 1974; EIA Notification 2006 (& amendments).
- **Clearances (s11–12): GOI:** widening of existing State Highway – **does not fall under any EC category** per EIA 2006. **World Bank:** classified **Category B**; cumulative impacts considered. **State Ground Water Board:** clearance for ground water extraction. **State pollution control board:** consent for establishment and operation of construction plants. **Forest clearance** from Forest dept for tree cutting.
- **With vs without project (s13–14):** without – poor connectivity/pavement, high travel time, pollution, restricted access to markets, medical, education; no impact on ecology/social. With – improved connectivity, reduced travel time and pollution, increased trade; **felling of 186 trees** (compensatory afforestation); **03 structures (encroachments)** affected, no common property resources affected.

Public consultation L19·s15–21

- Consultation (s15): **Antroli Village** (chainage 98+300, **18** participants) and near starting point **Gumma Village** (95/000, **16** participants). Typical issues: drainage problem; road safety; bus stop; dust suppression; speed breaker; relocation of structures; junction improvement. Attendance sheet dated 22-09-2019 (s16).

Issue raised	Redressal (s17–21)
Loss of income restoration option	Adequate compensation
Road safety	Traffic calming at intersections, crossings; **hazard markers, RRPM (Raised Reflective Pavement Marker)**, road marking; **crash barrier** throughout alignment
Land acquisition	Compensation as per **RAP** (Resettlement Action Plan)
Flood problem / drainage	**Total 38 culverts**; drainage throughout alignment
Loss of trees	Compensatory forestation **two trees for each tree cut** [s20]; local species
Impact on health; utilities and basic infrastructure; employment	Further study of bypass impact on health; roadside plantation to screen emissions; utilities relocated under project cost; locals preferred for construction employment

- **Devices (s18–19):** hazard markers = signs showing upcoming bridge piers/traffic islands (precautionary). RRPM maintain traffic direction, dictate pavement limit, reflective for night visibility. **Crash barriers** = steel structures along road limiting vehicles going off road; especially useful on hilly roads (valley side).

Impacts & mitigation L19·s22–33

- **Major impacts (s22):** soil erosion & landslides; deforestation; flooding & drainage; air pollution; noise pollution; land use change; road safety.

Impact	Mitigation (design)	Mitigation (implementation)
Soil erosion (biggest challenge on hilly terrain)	Identify erodible soils; treat embankment slopes and stream inlet/outlet	Close control of timing of embankment treatment after earthwork
Deforestation	Minimize via alignment selection for ROW	Control tree cutting; afforestation in **1:3 ratio** [s28]
Flooding/drainage	38 culverts; drainage throughout	Keep channels clear before monsoon
Air pollution	Wider road with paved shoulders	Work areas/camp away from sensitive receptors; modern equipment; dust suppression (water sprinkling)
Noise	Flexible pavement; vegetation barrier	Work areas away; well-maintained equipment; acoustic enclosures for DG sets
Land use	Shortest alignment where social disruption outweighs farmland loss	Reinstate working areas
Road safety	Median for all sections (avoid head-on collision); traffic calming near junctions/villages; pedestrian crossings; crash barriers; high-standard signs, hazard markers, markings	Safe traffic management methods

- **Slope stabilization (s24–27): rock anchoring/nailing** – long anchor bolt stabilizing rock excavations, tunnels/rock cuts, transfers load from unstable exterior to confined interior; **hydro-seeding** – mixture of water, seed, wood fiber, soil stabilizer; **bamboo net plantation** – for dumping sites, natural barrier; **hedge brush layer** – local shrubs on slope, over valley-side embankment (except built-up area, bridges, culverts).

Monitoring L19·s34–35

Impact	Parameters	Standards	Frequency
Air	CO, NOx, SPM, RPM, SO_2	Air (Prevention & Control of Pollution) Rules, CPCB, 1994	Quarterly
Water	Physical, chemical, biological	IS 2296 (1982) inland surface waters; IS 10500 (1991) drinking water	Quarterly
Noise	Noise level (dB)	MoEF&CC Noise Rules, 2000	Quarterly
Afforestation	Survival rate	—	Quarterly
Soil	Pb (lead), SAR (sodium adsorption ratio), oil & grease	Threshold per contaminant set by IRIS database of USEPA	Half-yearly

- **Conclusion (s36):** economically very beneficial with little negative social impact; soil erosion and landslides = key concerns; proper mitigation → little environmental impact.

Important Distinctions to Remember

- GOI: not in any EC category (widening of existing State Highway); World Bank: **Category B**.
- Tree compensation stated as **2 per tree cut** (public-meeting redressal, s20) and **1:3** afforestation ratio (impact table, s28) – slide figures differ; keep both.
- Hazard markers (signs) vs RRPM (reflective pavement markers) vs crash barriers (steel).
- Monitoring: air/water/noise/afforestation quarterly; soil half-yearly.

Image slides read: s8, 15, 16, 23, 29, 34, 35. s14 text table read from extraction; s17 "Redressal" text from extraction.

Lecture 20 — EIA Case Study-IV: Guwahati airport integrated terminal (36 slides)

Project L20·s3–13

- New Integrated Terminal Building at **Lokpriya Gopinath Bordoloi International Airport (LGBI) / Guwahati International Airport**; proponent **Airport Authority of India (AAI)**; EIA by "ABC Techno Labs India Private Limited".
- **AAI (s4):** constituted by an Act of Parliament in **1995**; creates, upgrades, maintains, manages civil aviation infrastructure; manages **137 airports**; air navigation services over **2.8 million square nautical miles** of airspace.
- **Guwahati airport (s5–7):** major airport in North-East India, Assam; inter-region hub; suits medium aircraft (**B-767/A-310**); before project: **75 flight movements/day, 525/week**; saturated; land **528.225 acres**. **No alternative site** considered – existing terminal saturated; site within airport (s8).
- **Planned project (s9):** centrally air-conditioned modular terminal with multilevel car parking; designed for **2900 domestic and 200 international passengers at a time**; **Level of Service "B"** per IATA initially, **"C"** at saturation. **LOS (s10):** A Excellent comfort; B High; C Good; D Adequate; E Inadequate; F Unacceptable — depends on space per passenger and maximum waiting time; IATA changes standards from time to time.
- **Benefits (s11):** better passenger facilities; tourism, trade, commerce; revenue; employment; business and industrial opportunities.
- **EC requirement (s12):** per EIA 2006 all projects incl. airstrips for commercial use need EC from MoEF&CC; **Category A**; air strips without bunkering/refueling facility and/or Air Traffic Control are **exempted**. **ICAO norms (s13):** reducing noise; land use planning; reducing emissions at source; operational mechanisms; market-based options (emissions permit trading, charges, voluntary agreements).
- **Community consultation (s14):** group discussions; issues: employment, information flow, health & safety, social welfare; people expect preference for locals in construction jobs.
- **Impact rating matrix (s15): Negligible** – change comparable to natural variation; **Minor** – detectable but not significant; **Moderate** – insignificant; amenable to mitigation; should be mitigated where practicable; **Major** – significant; amenable to mitigation; **must be mitigated**.

Impacts and mitigation L20·s16–29

Component	Mitigation / facts
Topography (s16)	Design minimizes earth filling; borrow only from approved borrow area; rehabilitate borrow area
Land use (s17)	Minimum land clearing; minimum cut and fill; debris collected and disposed in sound manner
Water resource (s18)	Total water requirement **1010 kld (fresh water 710 kld)**; reduce consumption; sewage treated in **STP**; treated water reused in HVAC, flushing, greenery, landscaping
Water quality (s19)	Wastewater generation in operation **808 kld**; STP; avoid fuel/lube spillage (store on concrete floor); solid waste disposal; regular testing
Air quality (s20–21)	Emission rates (g/s) – Aircraft: SO_2 5.62, NOx 4.80, CO 47.39, HC 70.30; DG set: SO_2 2.92, NOx 3.26, PM 0.54; Vehicular: NOx 1.87, CO 3.27, HC 0.85, PM 0.32. Rating: adverse, long term, localized, likelihood high, severity **medium**, significance **moderate**. Mitigation: ICAO standards; **30 m stack height** of DG sets; no traffic congestion; vehicles with PUC; greenery
Noise (s22–23)	Noise level vs area of influence: 80 dB – 0.07 km^2; 75 – 0.21; 65 – 1.94 (all within airport boundary); **55 – 5.32 km^2** (about 2.4 km^2 outside); **50 – 8.63 km^2** (about 6.7 km^2 outside). Rating: adverse, long term, localized, likelihood high, severity **low**, significance **low**. Mitigation: no congestion; acoustic enclosure for DG sets; soundproof terminal; ground staff earplugs; green belt/landscaping/boundary as barrier; plantation in nearby settlements
Terrestrial ecology (s24)	**No tree felling** required; more landscaping; indigenous species planted
Deepor Beel bird sanctuary (s25–26)	Ramsar site **3 km** from project; **not in funnel area** of runway (imaginary flight path, up to ~15°); noise modelling: aircraft noise impact **negligible**. Precaution: landscaping/green belt toward sanctuary; implement Chief Wildlife Warden recommendations
Employment & growth (s27)	Boost in tourism/commercial activity; direct/indirect employment; revenue

Energy conservation (s28–29)	Designed for **GRIHA 4 star** rating; energy-efficient materials/glass; **LED lamps instead of GLS**; efficient HVAC; **5-star BEE** equipment; microprocessor-based **BMS**; automatic lighting control; **540 kW solar plant**; about **30% energy saved**

Monitoring; reporting; risks L20·s30–33

Component	Stage	Parameters	Location	Frequency
Air	Construction & operation	PM2.5, PM10, SO_2, NO_2, CO	Project site	24 hourly, once in season
Water quality	Construction (IS 10500:2012); operation (IS 10500:2013)	As per IS	Project site	Half-yearly
Noise	Construction; operation	Day & night noise dB(A)	4 locations on airport boundary; arrival & departure location, boundary	Once in season
Treated wastewater from STP	Operation	pH, TDS, TSS, BOD, COD, oil & grease	Outlet of STP	Once in month

- **Reporting (s31):** compliance monitoring and progress reports **monthly** during implementation; operation-stage reports half-yearly or annually; ensure actions per contract; photographic records.
- **Risks (s32):** HSD (high speed diesel) storage for DG sets in underground tanks (leakage); fire in **ATF** (aviation turbine fuel) tanks; aircraft crash while landing/take-off; leakage of flammable materials followed by fire; natural calamities (earthquake, cyclone, high winds). External threats: agitation/forced entry; sabotage; air raids; bomb threat. **Flammable spill provisions (s33):** non-flammable absorbent material ready; ignition sources quickly removed; well ventilated; spread of spill limited.
- **Conclusion (s34):** all impacts assessed with mitigation; periodic monitoring; risks included.

Important Distinctions to Remember

- Airport = **Category A** (EIA 2006); airstrips without refueling/ATC exempted. LOS B initially → C at saturation.
- Air impact: severity medium, significance moderate; noise: severity low, significance low (both adverse, long term, localized, high likelihood).
- Rating matrix: negligible / minor / moderate (should be mitigated) / major (must be mitigated).
- Deepor Beel (Ramsar, 3 km, not in funnel area, negligible noise). 30 m DG stack; 540 kW solar; ~30% energy saved; GRIHA 4 star.

Image slides read: s6, 15, 20, 22, 30. Water numbers (1010/710/808 kld) from slide text.

Part 3 — Rapid-Revision Sheet (Lectures 11–20)

Memory aid only — full facts are in Part 2.

EIA basics (L11–L13)

- **EIA (Sadler 1996):** identifying, predicting, evaluating, mitigating biophysical, social and other effects **prior to major decisions**. Middle path (**Madhyam-Pratipada**); balances development vs environment.
- **Core values:** Integrity, Utility, Sustainability. **Purpose:** facilitate decision-making; aid sustainable development; instrument for sustainable development.
- **Types:** Rapid (1 season) / Comprehensive (4 seasons); Regional, Sectoral, Strategic (SEA = EIA of PPP), Project-level, LCA. **EIA** end-of-cycle, reactive vs **SEA** early, proactive.
- **Impact classes:** beneficial/detrimental; reversible/irreversible; repairable/irreparable; short/long term; local/regional/national.
- **Stages:** Project proposal → Screening → Scoping → Baseline → Impact prediction/assessment → Mitigation → EIS → Review (public consultation) → Licensing → Follow-up/monitoring.
- **Screening:** is EIA needed? (threshold & impact criteria; inclusion/exclusion lists; categories A/B/C). **Scoping:** boundaries → **TOR**. **Baseline:** current status. World Bank A/B/C; C e.g. PMGSY village roads.
- **Mitigation:** prevent / limit severity & duration / compensate. Hierarchy **Avoid** → **Mitigate** → **Restore** → **Offset**. IMP elements: mitigation measures, monitoring, capacity development, implementation schedule & cost, integration. **EIS** = final stage; review by MoEF (India); licensing = outcome of review; follow-up checks accuracy of predictions.

Key dates / names

Item	Fact
NEPA	1969 (US)
Espoo Convention	1991; first multilateral EIA treaty; in force 1997
Rio Declaration	1992; Principle 17
India EIA	1976-77 start; 1994 notification; 1997 public hearing compulsory; Sept 2006 notification; March 2020 draft
Project Tiger	1973
Bhopal	1984; MIC gas; 3787 died; 574336 affected
National Forest Policy	1988 (one third tree cover); Arts 48A, 51A(g), 21
Policies	Conservation Strategy 1992; Abatement of Pollution 1992; Water Policy 2012; Environment Policy 2006

Methodologies (L14)

Method	Key words
Ad hoc	intuitive, qualitative, expert team, screening; rank alternatives 1–4
Checklist	screening & scoping; simple / descriptive / scaling(weights: air 40, noise 15, wildlife 10, local pop 20, water 15 = 100)
Matrix	environmental components × project actions
Network	impact chains; branches; Grand Network Index = 106.2 (33.8 + 51.2 + 21.2)
Overlay	maps at project scale; transparencies → GIS
Simulation	micro, experimental, single activity (EV vs fossil vehicles)
Cost-benefit (SCBA)	public sector; monetary terms; World Bank; 5 stages incl. shadow prices
Index / EPI	predefined standards; EPI: 32 indicators, 11 categories, 2 objectives; distance-to-target (0–100); Env. Health 40 / Ecosystem Vitality 60

EIA in India (L15) and abroad (L16)

Item	2006	Draft 2020

Categories	A (MoEF&CC-EAC), B (SEIAA)	A, B1, B2
Baseline data	yearly	one season (not for river valley)
Monitoring	6-monthly, third party	yearly, by proponent
Highway A	> 30 km, ROW > 20 m, > 1 State	NH/expressway new; expansion > 100 km, ROW > 70 m

- 2006 numbers: SEIAC 3 members (reconstituted every 3 yrs); Form 1A 93 parameters/10 categories; ToR within **60 days**; recommendations to MoEF&CC in **15 days**; appraisal within **60 days**; compliance reports **1 June & 1 Dec**; B→A if within **10 km** of protected areas etc. Railways not under 2006 notification. Port A: > 5 million TPA.
- Global: SEA integrated (Kenya, Tanzania) / separate laws (Indonesia, Canada) / sectoral (NZ). Parameters: environment-centred, development-centred, **hybrid (India, EU)**. EU 2 categories; Peru 3 (EIS, EIA-SD, EIA-D). Canada 20-day comment, cumulative effects. Kenya practitioner at screening. Splitting (Spain, UK); exemptions (Nigeria, Indonesia).

Case studies (L17–L20)

Case	Key numbers / facts
MAHSR (L17)	508 km; 12 stations; 350 kmph; Shinkansen; 6.5→2.5 hr; NHSRCL (SPV, 12 Feb 2016); JICA soft loan 12 bn USD, 0.1%, 50 yr incl. 15-yr grace; cost INR 1.08 lakh crore; JICA Category A; viaduct 2.8 ha/km vs embankment 3.2 ha/km vs conventional 7.5; forest 138.3269 ha (mangrove 32.3902); 60000 trees; completion 2028; SEMU executes monitoring
Xiangjiang (L18)	ADB Category A; 50 km reach; 90 MW; 358.2 MkWh; upstream site chosen; 300 questionnaires/100%/95% agree; hearing 4 Aug 2010; groundwater rise 0.5–2.1 m; fish pass + hatchery; new vs old grievance mechanism
NH-707 (L19)	Gumma–Fediz; NHIIP/World Bank Category B; 2500–3500 m; 8 landslide points; 38 culverts; 186 trees; hazard markers, RRPM, crash barriers; monitoring quarterly (soil half-yearly)
Guwahati airport (L20)	AAI (1995, 137 airports); LGBI; Category A; LOS B → C; 2900 domestic + 200 international pax; water 1010 kld (fresh 710); wastewater 808 kld; 30 m DG stack; 540 kW solar; ~30% energy saved; Deepor Beel 3 km (Ramsar), negligible noise

Part 4 — Practice MCQs (3 per lecture = 30)

Q1 (L11). Which is described as the first multilateral EIA treaty?
A. Rio Declaration B. NEPA C. Espoo Convention D. Environment (Protection) Act

Q2 (L11). A Comprehensive EIA collects baseline data for:
A. one season B. two seasons C. three seasons D. four seasons

Q3 (L11). According to the EIA timeline, public hearing was made compulsory in India in:
A. 1991 B. 1994 C. 1997 D. 2006

Q4 (L12). A project likely to have minimal or no adverse impacts, needing no further EA action beyond screening, is World Bank:
A. Category A B. Category B C. Category C D. Category D

Q5 (L12). For a highway, the "net project area" is:
A. area up to buildings B. area affected by noise and air pollution C. right of way of highway D. entire district

Q6 (L12). In the Environmental Clearance process flow, the public hearing is arranged by:
A. SPCB B. MoEF&CC C. SEIAA D. NGT

Q7 (L13). The mitigation hierarchy steps in order are:
A. Mitigate, Avoid, Restore, Offset B. Avoid, Mitigate, Restore, Offset C. Avoid, Restore, Mitigate, Offset D. Offset, Restore, Mitigate, Avoid

Q8 (L13). Project Tiger was launched in:
A. 1973 B. 1984 C. 1991 D. 2006

Q9 (L13). The Bhopal Gas Tragedy figures on the slide are:
A. 574336 died and 3787 affected B. 3787 died and 574336 severely affected C. 3787 died and 5743 affected D. 11 died and 1000 sick

Q10 (L14). The EPI uses ___ indicators grouped into ___ issue categories:
A. 11; 32 B. 32; 11 C. 40; 60 D. 20; 16

Q11 (L14). In the network-method example, the Grand Network Index is:
A. 33.8 B. 51.2 C. 21.2 D. 106.2

Q12 (L14). In the scaling checklist example, the weightage given to Air Quality is:
A. 10 B. 15 C. 20 D. 40

Q13 (L15). Under EIA notification 2006, Category B projects are appraised by:
A. MoEF&CC-EAC B. SEIAA C. District Level Committee D. NGT

Q14 (L15). Form 1A requires details on how many parameters in how many categories of impact?
A. 93 parameters in 10 categories B. 10 parameters in 93 categories C. 18 parameters in 9 categories D. 60 parameters in 6 categories

Q15 (L15). Under the 2006 scoping process, if ToR is not conveyed within ___ days, the provided ToR is assumed final:
A. 15 B. 30 C. 60 D. 90

Q16 (L16). In Canada, the public is invited to submit comments for a period of:
A. 10 days B. 20 days C. 30 days D. 60 days

Q17 (L16). In Peru, Category-III projects need to undergo:
A. only an Environmental Impact Statement B. a Semi-detailed study (EIA-SD) C. a Detailed study (EIA-D) D. no study

Q18 (L16). Which country's measure against project splitting is "consider all the associated developments part of a project"?
A. Portugal B. Sweden C. Austria D. Germany

Q19 (L17). The Japanese soft loan for MAHSR is 12 billion USD at ___ interest, repaid over ___ including a ___ grace period:
A. 1%; 30 years; 5-year B. 0.1%; 50 years; 15-year C. 0.1%; 15 years; 50-year D. 10%; 50 years; 15-year

Q20 (L17). MAHSR reduces travel time from:
A. 6.5 hrs to 2.5 hrs B. 8 hrs to 1.25 hrs C. 12 hrs to 6 hrs D. 2.5 hrs to 6.5 hrs

Q21 (L17). Which unit executes the mitigation plans and monitoring for MAHSR?
A. NHSRCL board B. SEMU C. JICA D. DTEPA

Q22 (L18). For the Xiangjiang barrage, the location selected was:
A. Upstream B. Middle C. Downstream D. none

Q23 (L18). The hydropower station installed capacity in the Xiangjiang project is:
A. 9 MW B. 50 MW C. 90 MW D. 358.2 MW

Q24 (L18). In the group investigation, 300 questionnaires were handed out; what percentage of residents agreed the project is conducive to regional economic development?
A. 50% B. 75% C. 95% D. 100%

Q25 (L19). Elevation of the NH-707 project area varies from:
A. 500 m to 1000 m B. 1000 m to 2500 m C. 2500 m to 3500 m D. 3500 m to 5000 m

Q26 (L19). The NH-707 project was classified by the World Bank as:
A. Category A B. Category B C. Category C D. not classified

Q27 (L19). Total number of culverts provided for flooding/drainage is:
A. 18 B. 28 C. 38 D. 93

Q28 (L20). The Guwahati terminal is designed for level of service ___ per IATA in the initial years:
A. A B. B C. C D. F

Q29 (L20). The solar power generation plant to be established at the airport is:
A. 54 kW B. 90 kW C. 540 kW D. 5400 kW

Q30 (L20). Deepor Beel Bird Sanctuary is located ___ from the project site:
A. 1 km B. 3 km C. 10 km D. 15 km

Answer key

Q	Ans	Source-based explanation
1	C	Espoo (1991) = first multilateral EIA treaty — L11 s19
2	D	Four seasons (monsoon, post-monsoon, winter, summer) — L11 s22
3	C	1997 — L11 s15
4	C	World Bank C — L12 s24
5	C	Net project area = right of way — L12 s10
6	A	"SPCB arranges public hearing" — L12 s8
7	B	Avoid, Mitigate, Restore, Offset — L13 s9
8	A	1973 — L13 s7
9	B	3787 died, 574336 severely affected — L13 s24
10	B	32 indicators into 11 issue categories — L14 s40
11	D	106.2 — L14 s27
12	D	Air quality weight 40 — L14 s19
13	B	B category done by SEIAA — L15 s13
14	A	93 parameters in 10 categories — L15 s19
15	C	Sixty days — L15 s23
16	B	20 days — L16 s18
17	C	EIA-D — L16 s9
18	C	Austria — L16 s36
19	B	0.1%, 50 years, 15-year grace — L17 s11
20	A	6.5 → 2.5 hrs — L17 s9
21	B	SEMU — L17 s35
22	A	Upstream location selected — L18 s19
23	C	90 MW — L18 s6
24	C	95% agree — L18 s29

25	C	2500 m to 3500 m — L19 s6
26	B	World Bank Category B — L19 s11
27	C	38 culverts — L19 s20, s29
28	B	Level B initially, C at saturation — L20 s9
29	C	540 kW — L20 s29
30	B	3 km — L20 s25

Part 5 — Coverage Audit (Lectures 11–20)

Method: each PDF's text layer was extracted and read slide by slide; every slide with little or no text (diagrams, tables as images, charts) was rendered and read visually. Red pen marks on slides were ignored. Slide content that is simply absent from the PDF is listed as such.

PPT filename	Total slides	Slide numbers covered in notes	Extraction issues
lec 11.pdf	47	1–45 (46 references, 47 thank-you)	Image slides (10, 11, 15, 21, 24, 34, 39, 42) read from renders. s4, s26 photo/map: captions only.
lec 12.pdf	51	1–49 (50 references, 51 thank-you)	Diagram slides read from renders. **s45 has only a title** (content absent). EU checklist item 10 missing from extracted text (s26).
lec 13.pdf	43	1–41 (42 references, 43 thank-you)	**Content absent in PDF:** s26–28 (EIS formats), s30 (review objectives 1–4 blank), s34 (steps for public participation), s37 (World Bank monitoring), diagrams on s38–39. Tiger table (s8) and shortcomings (s40) read from renders.
lec 14.pdf	45	1–43 (44 references, 45 thank-you)	**Content absent in PDF:** s18, s21, s24, s28, s30 (pros/cons slides and scaling-checklist features); s5 etc. text-extracted. EPI chart values (s41) read from render.
lec 15.pdf	49	1–47 (48 references, 49 thank-you)	Category tables (s15, 16, 33–35) read from renders because the text layer misaligned columns.
lec 16.pdf	40	1–38 (39 references, 40 thank-you)	Checklist/legal-text images (s5–7, 9, 10, 25) and s36 table read from renders.
lec 17.pdf	39	1–37 (38 references, 39 thank-you)	Map (s8), tables (s10, 14), grievance flowchart (s33) and EMoP diagram (s35) read from renders. s10 shows "6 times by 20250" as printed.
lec 18.pdf	42	1–40 (41 references, 42 thank-you)	Location-alternative tables (s15–19), impact flowchart/matrix (s20–21) and grievance diagrams (s34–35) read from renders. s20: upstream water-level rise figure illegible.
lec 19.pdf	38	1–36 (37 references, 38 thank-you)	s8, 15, 16 (consultation sheet: handwritten, not transcribed), 23, 29, 34, 35 read from renders. Tree-compensation ratios differ between s20 (2 per tree) and s28 (1:3).
lec 20.pdf	36	1–34 (35 references, 36 thank-you)	s15, 20, 22, 30 read from renders; s6 map labels only.

Totals: 10 files, 430 slides (47+51+43+45+49+40+39+42+38+36).

- **Check 3 (no outside info):** only slide wording used; abbreviations expanded only where the slides expand them.
- **Not recoverable:** the slides marked "content absent in PDF" above — please check your original PPTs.
- **Not done:** I did not view the finished PDF page by page.